Finite-sample conservatism of the Stringer bound: sharpness at $n = 2$ and certified guarantees at conventional confidence levels

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Abstract

The Stringer bound is widely used to evaluate monetary-unit samples in auditing, but its finite-sample coverage is not known in general. For sample size $n = 2$, we prove that the binomial-factor bound is conservative for every distribution on $[0, 1]$ and every nominal confidence level. The result is sharp: the supremum of noncoverage is α and is not attained. At 90%, 95%, and 99% confidence, exact Bernstein certificates extend binomial-factor conservatism through $n = 6$. For the Poisson factors used in audit practice, a corrected simultaneous-survival-band argument and exact rational certificates establish conservatism through $n = 8$, $n = 11$, and $n = 20$, respectively. For every sample size and confidence level, scalar calibration of the Poisson factor curve yields a distribution-free valid modified rule. At nominal confidence at least $2/e \approx 73.6\%$, the multiplier two is valid uniformly in the sample size. Retaining the first 100 Poisson crossing probabilities gives certified uniform multipliers 1.53, 1.44, and 1.33 at 90%, 95%, and 99%, respectively. An anchored variant leaves the all-zero-sample result unchanged. A separate all-sample-size rule reports the maximum of Stringer and a distribution-free bounded-mean limit. We also give explicit finite-sample counterexamples at selected confidence levels between 30% and 37% and sample sizes $n = 50, 100, 200, 400$; exact rational certificates show coverage below nominal. In searches at 50%, 90%, and 95%, no violation was found, and the smallest coverage found agreed with nominal to within 10^{-10}. Finally, we identify an index shift in a previous finite-sample certification. These results give sharp distribution-free theory at $n = 2$, prove conventional-level coverage over new finite ranges, provide valid all-sample-size reporting alternatives, and narrow the unresolved regime for ordinary Stringer.

Keywords: Stringer bound; finite-sample coverage; bounded means; monetary-unit sampling; Poisson confidence factors; simultaneous bands; Clopper–Pearson limits.

1 Introduction

Monetary-unit sampling (MUS) draws n dollar units at random from an audited population. The *taint* of a drawn unit is $T \in [0, 1]$, the overstated fraction of the account item associated with that dollar. With taints $t_{(1)} \geq \cdots \geq t_{(n)}$ arranged in nonincreasing order, the Stringer bound [21] at nominal confidence $1-\alpha$ is

$$\mathrm{SB} \;=\; p_n(0) \;+\; \sum_{j=1}^{n} \bigl(p_n(j) - p_n(j-1)\bigr)\, t_{(j)}, \tag{1}$$

where $p_n(j)$ is the upper Clopper–Pearson limit for j successes in n trials, i.e. the root of $\sum_{i=0}^{j} \binom{n}{i} p^i (1-p)^{n-i} = \alpha$, with $p_n(n) = 1$. Audit guidance also tabulates Poisson-based confidence factors for MUS evaluation [1, Appendix C]. Proposition 9 below proves that these factors dominate the binomial factors, at every sample size, whenever nominal confidence exceeds $1 - e^{-1} \approx 63.2\%$. The method is commonly justified by the conjecture that

$$\mathbb{P}\bigl(\mathrm{SB} \ge \mu(F)\bigr) \;\ge\; 1-\alpha \qquad \text{for all } n \text{ and all } F \text{ on } [0,1], \tag{2}$$

where $T_1, \dots, T_n$ are drawn independently from the common distribution F, with mean $\mu(F)$. This is the standard i.i.d. model used in published analyses of the bound. A National Research Council panel noted that the formulation of the Stringer bound had not been satisfactorily explained [13]. The principal prior results are as follows. Bickel [5] proved (2) for $n = 1$, gave the lower bound $(1-\alpha)^{n+1}$ for $n \ge 2$ under conditions on F, and derived the expansion $\mathrm{SB} = \bar{T} + c(F) z_{1-\alpha} n^{-1/2} + o_p(n^{-1/2})$ with $c^2(F) \ge \mathrm{Var}(T)$. Pap and van Zuijlen [15] established an almost-sure version of this expansion and concluded that the bound is asymptotically conservative for $\alpha \in (0, \frac{1}{2}]$ and asymptotically *anti*-conservative for $\alpha \in (\frac{1}{2}, 1)$. In [14], they exhibited non-conservatism for uniform-type taints at sub-50% confidence using exact recursions. De Jager, Pap, and van Zuijlen [7] proved that, for taint distributions supported on $\{0, 1\}$, the bound reduces to Clopper–Pearson (where (2) holds exactly) and that Stringer's coefficients are minimal with that property. A 2021 survey of confidence bounds for bounded means categorized the Stringer bound among methods conjectured to have guaranteed coverage and described its coverage for $\alpha < 0.5$ as unknown [18, p. 8569]. Vlassis and Thomas [20] recently proved the finite-sample validity of Gaffke's distribution-free mean test [8]. Inversion for observations in $[0, 1]$ gives an upper confidence limit that is a quantile of a uniform Dirichlet average [11, 12]. This development supplies the valid comparison bound used for our $n = 3$ through $n = 6$ results.

This paper makes five contributions.

(1) A theorem at $n = 2$ (Section 2). The conjecture (2) holds at $n = 2$ for every F and every $\alpha \in (0, 1)$, including $\alpha > 1/2$. Thus the known asymptotic failures do not occur at $n = 2$, and the result is sharp. To our knowledge, this is the first distribution-free finite-sample proof for $n = 2$.

(2) Conventional confidence levels from $n = 3$ through $n = 6$ (Section 3). At nominal confidence 90%, 95%, and 99%, the binomial Stringer bound pointwise dominates the valid Gaffke upper limit. The comparison is proved by exact rational Bernstein certificates for the relevant uniform-simplex cap volumes. It yields distribution-free finite-sample conservatism at each of these levels for $n = 3, 4, 5, 6$. A directed-interval Bernstein certificate handles the five-dimensional case. The same comparison suggests a direct all-sample-size implementation safeguard (Section 5): pre-specifying the maximum of Stringer and the valid Gaffke limit retains Stringer whenever it is larger and has distribution-free coverage without assuming the unresolved conjecture. A dimension-free cap lemma and an analytic Clopper–Pearson comparison further prove a directly checkable zero-uplift region for every n whenever nominal

confidence is at least 75%: binomial Stringer dominates Gaffke whenever it is at least the largest observed taint.

(3) Guarantees for the factors used in practice (Section 4). Whenever nominal confidence exceeds $1-e^{-1} \approx 63.2\%$, the Poisson factors dominate the binomial factors coordinatewise for every n. A summation-by-parts identity then shows that the Poisson-factor Stringer bound is pointwise no smaller on every observed sample. A separate corrected simultaneous-band argument, certified by exact rational boundary-crossing calculations, proves Poisson-factor coverage for every $n \le 8$ at 90%, every $n \le 11$ at 95%, and every $n \le 20$ at 99% confidence. The same argument yields an all-sample-size calibrated rule that retains the shape of the ordinary Poisson factor curve: multiply every factor, equivalently the complete bound, by one precomputed scalar. We prove validity for every n and give exact rational certificates for representative audit sample sizes. An anchored variant fixes the zero-taint factor and scales only the error increments. Capping the calibrated factors at one gives pointwise no-larger valid versions. A sample-size-uniform analytic bound shows that doubling the Poisson curve is valid whenever nominal confidence is at least $2/e \approx 73.6\%$; at the three conventional levels, a finite-prefix refinement gives substantially smaller choices.

(4) Exact counterexamples at low confidence (Section 6). Thirty-three certified three-point distributions, each with two nonzero taint values and an atom at zero, give counterexamples at selected nominal confidence levels between 30% and 37%, for sample sizes $n = 50, 100, 200,$ and 400. Their coverage is evaluated in exact rational arithmetic with margin certificates. In searches at 50%, 90%, and 95%, no violation was found. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10}.

(5) Reassessment of a previous certification (Section 7). A proposed finite-sample certification [6] contains an index shift in its band-containment argument. We give hand-checkable examples for which the stated containment inequality fails and examine the corrected event.

2 Finite-sample conservatism at $n = 2$

For $n = 2$, the factors are explicit: $p_2(0) = 1 - \sqrt{\alpha}$, $p_2(1) = \sqrt{1-\alpha}$, $p_2(2) = 1$. Write

$$\beta = \sqrt{\alpha}, \quad \gamma = \sqrt{1-\alpha} \quad (\beta^2 + \gamma^2 = 1), \qquad A = \beta + \gamma - 1, \quad B = 1 - \gamma,$$

so $A, B > 0$ and $A + B = \beta$.

Theorem 1. *Let $\alpha \in (0,1)$ and let T_1, T_2 be i.i.d. with common distribution F on $[0,1]$ and mean μ. Then $\mathbb{P}(\mathrm{SB} \ge \mu) \ge 1 - \alpha$. Moreover $\sup_F \mathbb{P}(\mathrm{SB} < \mu) = \alpha$, and the supremum is not attained.*

2.1 Reduction

Substituting $U_i = 1 - T_i$, $w = 1 - \mu = \mathbb{E}[U]$, $m = \min(U_1, U_2)$, $M = \max(U_1, U_2)$ into (1) and telescoping gives

$$\mathrm{SB} \;=\; 1 - \big(A\,m + B\,M\big), \tag{3}$$

so $\mathrm{SB} \ge \mu \iff Am + BM \le w$. The theorem is implied by the *wedge inequality*: for U_1, U_2 i.i.d. on $[0,1]$ with mean $w > 0$,

$$\mathbb{P}\big(Am + BM < w\big) \;\ge\; \gamma^2. \tag{4}$$

(If $w = 0$ then $U \equiv 0$ a.s. and $\mathrm{SB} = 1 \geq \mu$ almost surely.)

If $w > \beta$ the event $\{Am + BM \geq w\}$ is empty, since the left side is at most $A + B = \beta$; if $w = \beta$ it is exactly $\{U_1 = U_2 = 1\}$ (as $A, B > 0$), of probability $\mathbb{P}(U = 1)^2 \leq w^2 = \alpha$ by Markov's inequality.

Henceforth, assume $0 < w < \beta$. The proof has three steps. First, the rectangle lemma converts a hypothetical failure of the wedge inequality into pointwise lower bounds on the survival function G. Second, boundary rectangles relate values of G on opposite sides of u^*. Third, the potential inequality combines those bounds and forces the integral of G to exceed its prescribed value w, producing a contradiction.

2.2 Rectangle bounds

Let $F(x) = \mathbb{P}(U \leq x)$, $G = 1 - F$, and recall $w = \int_0^1 G$. Define

$$u^* = \frac{w}{\beta} \in (0,1), \qquad a_0 = \max\Big(0, \frac{w - B}{A}\Big), \qquad T^\dagger = \min\Big(\frac{w}{B}, 1\Big),$$

and note $a_0 < u^* < T^\dagger$.

Lemma 2 (Rectangle lemma)**.** *If $0 \leq a \leq b \leq 1$ and $Aa + Bb < w$, then $\mathbb{P}(Am + BM < w) \geq 2F(a)F(b) - F(a)^2$.*

Proof. On $\{U_1 \leq a, U_2 \leq b\} \cup \{U_1 \leq b, U_2 \leq a\}$ we have $m \leq a$, $M \leq b$, hence $Am + BM \leq Aa + Bb < w$ (using $A, B \geq 0$). By inclusion–exclusion (the intersection is $\{U_1 \leq a, U_2 \leq a\}$ since $a \leq b$) the union has probability $2F(a)F(b) - F(a)^2$. □

Assume for contradiction that (4) fails. Then

$$2F(a)F(b) - F(a)^2 < \gamma^2 \quad \text{whenever } 0 \leq a \leq b \leq 1,\ Aa + Bb < w. \tag{H}$$

Three families of admissible rectangles give pointwise bounds on G:

(i) Squares. For $t < u^*$, $(a,b) = (t,t)$ gives $F(t)^2 < \gamma^2$, i.e. $G(t) > 1 - \gamma = B$ for all $t \in [0, u^*)$.

(ii) Full-height rectangles (nonvacuous if and only if $w > B$). For $a < a_0$, $(a, 1)$ gives $1 - G(a)^2 < \gamma^2$, i.e. $G(a) > \beta$ on $[0, a_0)$.

(iii) Boundary pairs. Let $h(g) = \dfrac{\beta^2 - g^2}{2(1-g)}$ on $[0,1)$. For $b \in (u^*, T^\dagger)$ set $\bar{a}(b) = (w - Bb)/A \in (a_0, u^*)$. Every $a' < \bar{a}(b)$ is admissible with b; letting $a' \uparrow \bar{a}(b)$ and writing $g = G(\bar{a}(b)^-) < 1$, inequality (H) rearranges to

$$G(b) \ \geq \ h\big(G(\bar{a}(b)^-)\big) \qquad \text{for all } b \in (u^*, T^\dagger)$$

because $2(1-g) - \gamma^2 - (1-g)^2 = \beta^2 - g^2$. If $g = 1$ the bound $G(b) \geq 0$ is used instead, consistent with the convention $h_+(1) = 0$ below.

2.3 Potential function and budget

The function $\tilde{k}$ below is chosen to combine the contribution of $G(a)$ with the corresponding lower bound for $G(b(a))$ after the change of variables used below. Let $h_+ = \max(h, 0)$, extended continuously by $h_+(1) = 0$, and define

$$\tilde{k}(g) \;=\; g + \frac{A}{B}\, h_+(g), \qquad g \in [B, 1].$$

Lemma 3. *$\tilde{k}(g) \geq \beta$ on $[B, 1]$, with equality exactly when $g \in \{B, \beta\}$.*

Proof. Since $\beta^2 - B^2 = \alpha - (1-\gamma)^2 = 2\gamma(1-\gamma)$, we get $h(B) = B$, hence $\tilde{k}(B) = B + \frac{A}{B}B = A + B = \beta$; also $h(\beta) = 0$ so $\tilde{k}(\beta) = \beta$. Differentiating, $h'(g) = \frac{(1-g)^2-\gamma^2}{2(1-g)^2}$, so h has its unique critical point at $g = B$ (where $(1-g)^2 = \gamma^2$) and decreases on $(B, 1)$; hence $h \geq 0$ on $[B, \beta]$ (it decreases from $B > 0$ to 0) and $h < 0$ on $(\beta, 1)$. Further $h''(g) = -\gamma^2/(1-g)^3 < 0$: h, and therefore $\tilde{k} = g + \frac{A}{B}h$, is strictly concave on $[B, \beta]$. A strictly concave function with equal endpoint values β satisfies $\tilde{k} \geq \beta$ on $[B, \beta]$, with equality only at the endpoints. On $(\beta, 1]$, $h_+ = 0$ and $\tilde{k}(g) = g > \beta$. $\square$

Proof of Theorem 1. Assume (H). The substitution $t = b(a) = (w - Aa)/B$ is a decreasing bijection from $[a_0, u^*]$ onto $[u^*, T^\dagger]$, with $|dt| = (A/B)\, da$ (endpoints: $b(u^*) = u^*$; $b(a_0) = 1 = T^\dagger$ if $w \geq B$, $b(0) = w/B = T^\dagger$ if $w \leq B$). Using $w = \int_0^1 G \geq \int_0^{a_0} G + \int_{a_0}^{u^*} G + \int_{u^*}^{T^\dagger} G$, we bound the tail with (iii) (the left-limit $G(a^-)$ agrees with $G(a)$ for all but countably many a, so the integrals coincide):

$$\int_{u^*}^{T^\dagger} G = \frac{A}{B}\int_{a_0}^{u^*} G(b(a))\, da \;\geq\; \frac{A}{B}\int_{a_0}^{u^*} h_+\big(G(a)\big)\, da.$$

By (i), $G > B$ pointwise on $[a_0, u^*)$, so Lemma 3 applies under the integral:

$$w \;\geq\; \int_0^{a_0} G + \int_{a_0}^{u^*} \tilde{k}\big(G(a)\big)\, da \;\geq\; \beta a_0 + \beta(u^* - a_0) \;=\; \beta u^* \;=\; w,$$

where $\int_0^{a_0} G \geq \beta a_0$ by (ii). Equality is forced throughout; we exhibit a strict inequality in every case:

- If $a_0 > 0$: (ii) is pointwise strict on $[0, a_0)$, so $\int_0^{a_0} G > \beta a_0$.
- If $a_0 = 0$ and $G \neq \beta$ on a positive-measure subset of $(0, u^*)$: since $G > B$ there, $\tilde{k}(G) > \beta$ strictly (Lemma 3), so $\int_0^{u^*} \tilde{k}(G) > \beta u^*$.
- If $a_0 = 0$ and $G = \beta$ a.e. on $(0, u^*)$: for each $b \in (u^*, T^\dagger)$ choose $a' < \bar{a}(b)$ with $G(a') = \beta$ (possible, as such a' have full measure in a nonempty interval); the *strict* inequality (H) at the admissible corner (a', b) reads $(1-\beta)(1+\beta-2G(b)) < \gamma^2 = (1-\beta)(1+\beta)$, forcing $G(b) > 0$ for every $b \in (u^*, T^\dagger)$; hence $\int_{u^*}^{T^\dagger} G > 0$ and $w \geq \int_0^{u^*} G + \int_{u^*}^{T^\dagger} G > \beta u^* = w$.

Each case yields $w > w$, a contradiction. Thus (H) is impossible, proving (4) and the coverage statement.

Sharpness. Distributions supported on $\{0, u\}$, with $F = (1-q)\delta_0 + q\delta_u$, give failure probability q^2 when $q < \beta$ (the pair (u, u), since $\beta u > qu = w$) and additionally the mixed pairs when $q < B$ (since $Bu > qu$), totalling $1 - (1-q)^2$; letting $q \uparrow \beta$ and $q \uparrow B$ respectively yields failure probabilities approaching $\beta^2 = \alpha$ and $1 - \gamma^2 = \alpha$. Hence $\sup_F \mathbb{P}(\mathrm{SB} < \mu) = \alpha$.

Non-attainment. If some distribution F attained the supremum, then $\mathbb{P}(Am + BM > w) = \alpha$. Such a distribution must have $0 < w < \beta$, since the strict failure event is empty in the boundary regimes handled above. The coverage result just proved gives $\mathbb{P}(Am + BM < w) \geq \gamma^2$; since $\alpha + \gamma^2 = 1$, attainment would therefore require

$$\mathbb{P}(Am + BM < w) = \gamma^2, \qquad \mathbb{P}(Am + BM = w) = 0. \tag{5}$$

The rectangle bounds consequently hold with weak inequalities. Repeating the budget argument in that form forces equality at every step. Equality on the first interval gives $G = \beta$ almost everywhere on $(0, a_0)$, and equality in the potential bound permits $G \in \{B, \beta\}$ almost everywhere on (a_0, u^*). Because G is nonincreasing, there is a $c \in [a_0, u^*]$ such that its almost-everywhere level is β before c and B after c. Set $d = b(c) = (w - Ac)/B$. Equality in the tail comparison, together with $h_+(\beta) = 0$ and $h_+(B) = B$, gives $G(b(a)) = 0$ for almost every $a \in (a_0, c)$ and $G(b(a)) = B$ for almost every $a \in (c, u^*)$. Because b is decreasing, this is the corresponding switch at d on the other side of u^*. Equality in the discarded tail and right-continuity then force

$$G(x) = \begin{cases} \beta, & 0 \leq x < c, \\ B, & c \leq x < d, \\ 0, & d \leq x \leq 1, \end{cases} \qquad Ac + Bd = w. \tag{6}$$

The change of variables gives $c \leq u^* \leq d$. Every distribution in (6) places positive probability on the boundary $Am + BM = w$. If $0 < c < d$, it has masses A and B at c and d, so the mixed pair (c, d) has probability $2AB > 0$. If $c = 0 < d$, the atoms at 0 and d have masses $1 - B$ and B, so their mixed pair has positive probability. The remaining case $c = d$ forces $c = d = u^*$, and the pair (u^*, u^*) has probability $\beta^2 > 0$. Each case contradicts (5). Hence the supremum is not attained. □

Remark 4 (Relation to known inequalities)**.** The equal-weights case of (4), which bounds $\mathbb{P}(X_1 + X_2 \geq t)$ for nonnegative i.i.d. variables with a given mean, is the classical sharp inequality of Hoeffding and Shrikhande [9] (extended to general k in [10]); the wedge inequality is its order-statistic-weighted analogue, with genuinely unequal weights except at $\alpha = 16/25$, where $A = B$. The sharp-bound literature for order statistics and L-statistics under moment conditions (see [4]) concerns expectations or distribution functions of single order statistics, not mean-constrained tail probabilities of weighted combinations; we are not aware of a prior result implying (4). Other distribution-free upper bounds for bounded means include those of Anderson [2] and Gaffke [8]; the latter's finite-sample validity for independent variables was proved recently [20]. Those results are not used here, and the proof of Theorem 1 is self-contained.

3 Conventional confidence levels from $n = 3$ through $n = 6$

The recent validity proof for Gaffke's mean test [20] provides a second route to finite-sample coverage. For a real random variable X, write $Q_q(X) = \inf\{z : \Pr(X \leq z) \geq q\}$. Given n

observed taints $t_1, \dots, t_n$, define

$$G_\alpha(t_1, \dots, t_n) = Q_{1-\alpha}\left(D_0 + \sum_{i=1}^{n} t_i D_i\right), \qquad (D_0, \dots, D_n) \sim \text{Dirichlet}(1, \dots, 1).$$

This is the one-sided Gaffke–Learned-Miller–Thomas upper limit [8, 11, 12]; it has coverage at least $1-\alpha$ for independent observations from every distribution on $[0,1]$. We prove that Stringer is pointwise no smaller at three conventional confidence levels for $n=3$, $n=4$, $n=5$, and $n=6$.

3.1 The case $n = 3$

Theorem 5. *Let $n = 3$ and let T_1, T_2, T_3 be i.i.d. with common distribution F on $[0,1]$ and mean μ. If $\alpha \in \{0.10, 0.05, 0.01\}$, then*

$$\text{SB} \geq G_\alpha(T_1, T_2, T_3) \quad \textit{for every observed sample,} \qquad \Pr_F(\text{SB} \geq \mu) \geq 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 3$.

Proof. Put

$$a = 1 - p_3(2), \quad b = p_3(2) - p_3(1), \quad c = p_3(1) - p_3(0), \quad d = p_3(0).$$

For ascending sample taints $t_1 \leq t_2 \leq t_3$, $\text{SB} = at_1 + bt_2 + ct_3 + d$. The four coefficients are positive and sum to one. Unless $t_1 = 1$, affine equivariance permits subtraction of t_1 and division by $1 - t_1$. It therefore suffices to use the four knots $(0, x, y, 1)$, where $0 \leq x \leq y \leq 1$, and the Stringer threshold

$$s = bx + cy + d.$$

By exchangeability of the Dirichlet weights, pointwise domination of G_α is equivalent to

$$V(x,y) := \Pr_D\{xD_1 + yD_2 + D_3 > s\} \leq \alpha. \tag{7}$$

The uniform-simplex cap formula gives

$$V(x,y) = \begin{cases} \dfrac{(1-s)^3}{(1-x)(1-y)}, & s \geq y, \\ \dfrac{1}{1-y}\left\{\dfrac{(1-s)^3}{1-x} - \dfrac{(y-s)^3}{y(y-x)}\right\}, & x \leq s \leq y, \\ 1 - \dfrac{s^3}{xy}, & s \leq x, \end{cases} \tag{8}$$

with boundary values defined by continuity. The Clopper–Pearson equations imply

$$(a+b+c)^3 = \alpha, \qquad (b+c+d)^3 = 1 - \alpha, \qquad (a+b)^2\{3 - 2(a+b)\} = \alpha. \tag{9}$$

We verify (7) in the three regions of (8). If $s \leq x$, put $g = b + c + d = (1-\alpha)^{1/3}$. The exact factor enclosures certify $b, c \leq g/3$ at each of the three values of α. Weighted AM–GM therefore gives

$$\frac{s}{g} = \frac{b}{g}x + \frac{c}{g}y + \frac{d}{g} \geq x^{b/g} y^{c/g} \geq (xy)^{1/3},$$

and the third line of (8) is at most α.

For $s \geq y$, set $X = 1-x$, $Y = 1-y$, $L = 1-s = a+bX+cY$, and $A = 1-b-c = a+d$. The domain is the triangle with vertices

$$\left(\frac{a}{A}, \frac{a}{A}\right), \qquad (1,1), \qquad \left(1, \frac{a+b}{1-c}\right).$$

On that triangle the desired inequality is

$$P_A(X,Y) := \alpha XY - (a+bX+cY)^3 \geq 0. \tag{10}$$

After an affine map from the standard triangle, all degree-three Bernstein coefficients of P_A are positive except the coefficient at $(X,Y) = (1,1)$, which is zero by the first identity in (9). The coefficient signs are certified by exact rational intervals as described below.

It remains to consider $x \leq s \leq y$. Write $\rho = s - x \geq 0$ and $\tau = y - s \geq 0$. Solving for x, y shows that the (ρ, τ) domain is the quadrilateral with vertices

$$(0,0), \quad \left(\frac{d}{1-c}, 0\right), \quad (c+d, a+b), \quad \left(0, \frac{a}{1-b}\right).$$

Triangulate it along the diagonal from $(0,0)$ to $(c+d, a+b)$. Differentiating the middle line of (8) and cancelling common factors gives

$$\frac{\partial V}{\partial y} = \frac{Q(x,y)}{y^2(1-x)(y-x)^2},$$

where Q is a degree-five polynomial. On both triangles, every degree-five Bernstein coefficient of A^3Q is nonnegative. Three on each triangle are structural zeros, and all remaining coefficients are strictly positive. Hence V is nondecreasing in y at fixed x. Increasing y preserves both middle-region inequalities, so it is enough to take the limit $y \uparrow 1$.

At $y = 1$, put $q = 1 - x$. The constraint is $q \geq q_0 := a/(1-b)$, and direct simplification yields

$$V(x,1) = \frac{(a+bq)^2\{3q - (a+bq)(1+q)\}}{q^2}.$$

After mapping $[q_0, 1]$ to $[0,1]$, the degree-four Bernstein coefficients of

$$\alpha q^2 - (a+bq)^2\{3q - (a+bq)(1+q)\}$$

are positive except the coefficient at $q = 1$, which is zero by the final identity in (9). This completes (7).

For completeness, Table 1 reports the smallest positive Bernstein lower bound in each exact check. The factors are enclosed on a dyadic grid of width 2^{-120}; endpoint signs of their binomial CDF equations and every subsequent coefficient sign are evaluated with integer or rational arithmetic. The symbolic derivation starts from (8) and regenerates all coefficient formulas. Both the derivation and sign certificate are reproduced by the repository commands `make n3-formula-check` and `make n3-certificate-check`. Thus this is an exact computer-assisted proof, not a grid search. Finally, validity of G_α gives the stated coverage inequality. $\square$

α	region $s \geq y$	middle triangle 1	middle triangle 2	$y = 1$ boundary
0.01	1.04×10^{-7}	9.45×10^{-5}	1.37×10^{-9}	3.67×10^{-8}
0.05	1.63×10^{-5}	1.38×10^{-4}	7.52×10^{-8}	4.40×10^{-6}
0.10	1.45×10^{-4}	1.04×10^{-4}	3.27×10^{-7}	3.28×10^{-5}

Table 1: Smallest positive exact rational lower bound among the relevant Bernstein coefficients in the $n = 3$ certificate. Displayed values are decimal approximations; the complete rational bounds are in the machine-readable certificate.

3.2 The case $n = 4$

The same comparison can be carried out one dimension higher. The additional dimension changes the cap geometry substantially: after clearing boundary factors, two degree-six polynomials must be certified on six tetrahedra.

Theorem 6. *Let $n = 4$ and let $T_1, \dots, T_4$ be i.i.d. with common distribution F on $[0, 1]$ and mean μ. If $\alpha \in \{0.10, 0.05, 0.01\}$, then*

$$\mathrm{SB} \geq G_\alpha(T_1, \dots, T_4) \quad \textit{for every observed sample}, \qquad \Pr_F(\mathrm{SB} \geq \mu) \geq 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 4$.

Proof. Write $p_j = p_4(j)$ and put

$$a = 1 - p_3, \quad b = p_3 - p_2, \quad c = p_2 - p_1, \quad d = p_1 - p_0, \quad e = p_0.$$

For ascending taints $t_1 \leq t_2 \leq t_3 \leq t_4$, $\mathrm{SB} = at_1 + bt_2 + ct_3 + dt_4 + e$. The five coefficients are positive and sum to one. As in the $n = 3$ proof, affine equivariance reduces the sample knots to

$$(0, x, y, z, 1), \qquad 0 \leq x \leq y \leq z \leq 1, \qquad s = bx + cy + dz + e.$$

It is therefore enough to prove that

$$V(x, y, z) := \Pr_D\{xD_1 + yD_2 + zD_3 + D_4 > s\} \leq \alpha, \tag{11}$$

where $(D_0, \dots, D_4) \sim \mathrm{Dirichlet}(1, \dots, 1)$. Put

$$H_1 = \frac{(1-s)^4}{(1-x)(1-y)(1-z)}, \quad H_z = \frac{(z-s)^4}{z(z-x)(z-y)(1-z)}, \quad H_y = \frac{(y-s)^4}{y(y-x)(z-y)(1-y)}.$$

The uniform-simplex cap formula is

$$V = \begin{cases} H_1, & s \geq z, \\ H_1 - H_z, & y \leq s \leq z, \\ H_1 - H_z + H_y, & x \leq s \leq y, \\ 1 - \dfrac{s^4}{xyz}, & s \leq x, \end{cases} \tag{12}$$

with boundary values defined by continuity. The factor equations give

$$\begin{aligned}\alpha &= (1-e)^4,\\ \alpha &= (1-d-e)^3\{1+3(d+e)\},\\ \alpha &= (1-c-d-e)^2\{1+2(c+d+e)+3(c+d+e)^2\},\\ 1-\alpha &= (b+c+d+e)^4.\end{aligned} \tag{13}$$

First suppose $s \le x$ and set $g = b+c+d+e = p_3$. The exact factor enclosures certify $b, c, d \le g/4$ at each stated value of α. Weighted AM–GM gives

$$\frac{s}{g} = \frac{b}{g}x + \frac{c}{g}y + \frac{d}{g}z + \frac{e}{g} \ge x^{b/g}y^{c/g}z^{d/g} \ge (xyz)^{1/4}.$$

The last identity in (13) now makes the last line of (12) at most α.

For the other three regions, clear the positive denominators in (12). In the region $s \ge z$, the required residual is

$$P_A = \alpha(1-x)(1-y)(1-z) - (1-s)^4.$$

In the region $y \le s \le z$, use

$$\begin{aligned}P_B = {}&\alpha(1-x)(1-y)(1-z)z(z-x)(z-y)\\ &- (1-s)^4 z(z-x)(z-y) + (z-s)^4(1-x)(1-y).\end{aligned}$$

Exact polynomial division, using the second identity in (13), gives $P_B = (1-z)R_B$, where R_B has degree six. In the region $x \le s \le y$, the corresponding cleared residual is

$$\begin{aligned}P_C = {}&\alpha(1-x)(1-y)(1-z)z(z-x)(z-y)y(y-x)\\ &- (1-s)^4 z(z-x)(z-y)y(y-x)\\ &+ (z-s)^4(1-x)(1-y)y(y-x)\\ &- (y-s)^4(1-x)(1-z)z(z-x).\end{aligned}$$

The third identity in (13) gives $P_C = (1-z)(1-y)(z-y)R_C$, with R_C of degree six. Every extracted factor is nonnegative on the ordered-knot domain.

It remains to certify P_A, R_B, R_C. Define

$$\begin{gathered}r_0 = (0,0,0), \quad r_1 = \left(0, 0, \frac{e}{1-d}\right), \quad r_2 = \left(0, \frac{e}{1-c-d}, \frac{e}{1-c-d}\right),\\ r_3 = (t,t,t), \quad t = \frac{e}{1-b-c-d}, \qquad v_{001} = (0,0,1),\\ q_2 = \left(0, \frac{d+e}{1-c}, 1\right), \quad q_3 = (u,u,1), \quad u = \frac{d+e}{1-b-c},\\ v_{011} = (0,1,1), \qquad q_4 = \left(\frac{c+d+e}{1-b}, 1, 1\right).\end{gathered}$$

Intersecting $0 \le x \le y \le z \le 1$ successively with the planes $s = z$, $s = y$, and $s = x$ gives the following tetrahedralizations:

$$\begin{aligned} s \ge z: &\quad [r_0, r_1, r_2, r_3], \\ y \le s \le z: &\quad [r_3, r_2, r_1, v_{001}],\ [r_3, q_2, q_3, v_{001}],\ [r_3, q_2, r_2, v_{001}], \\ x \le s \le y: &\quad [r_3, q_2, r_2, v_{011}],\ [q_3, r_3, q_4, v_{011}],\ [q_3, r_3, q_2, v_{011}]. \end{aligned}$$

The listed tetrahedra meet only on common faces and cover the respective convex polytopes; this also follows directly by slicing at the common diagonals $[r_3, v_{001}]$ and $[r_3, v_{011}]$.

Map each tetrahedron affinely to the standard tetrahedron. The exact symbolic calculation finds one structural-zero coefficient in the degree-four Bernstein expansion of P_A. For R_B, the three degree-six expansions have respectively 10, 5, and 10 structural zeros; for R_C, they have 5, 10, and 10. These coefficients vanish identically over $\mathbb{Q}(b, c, d, e)$. Every remaining coefficient has a strictly positive exact rational interval enclosure at each stated level. Table 2 reports the smallest positive lower bounds by region. Factor endpoints lie on a dyadic grid of width 2^{-120}, their binomial-CDF signs are checked with integer arithmetic, and all polynomial, vertex, and Bernstein calculations thereafter use exact rational arithmetic. The symbolic and numerical layers are regenerated by `make n4-structure-check` and `make n4-certificate-check`. Hence $P_A, R_B, R_C \ge 0$, and (11) follows in all four regions. Validity of G_α then proves the coverage statement. $\square$

α	region $s \ge z$	region $y \le s \le z$	region $x \le s \le y$
0.01	3.10×10^{-10}	4.78×10^{-14}	5.84×10^{-12}
0.05	3.47×10^{-7}	2.66×10^{-10}	4.60×10^{-9}
0.10	7.74×10^{-6}	1.18×10^{-8}	7.93×10^{-8}

Table 2: Smallest positive exact rational lower bound among the relevant Bernstein coefficients in the $n = 4$ certificate. Displayed values are decimal approximations; the exact bounds and all structural-zero indices are machine-readable.

3.3 The case $n = 5$

At $n = 5$ the normalized knot domain is four-dimensional and all five cap regions require polynomial certificates. The exact computation is larger, but the comparison principle is unchanged.

Theorem 7. *Let $n = 5$ and let $T_1, \ldots, T_5$ be i.i.d. with common distribution F on $[0, 1]$ and mean μ. If $\alpha \in \{0.10, 0.05, 0.01\}$, then*

$$\mathrm{SB} \ge G_\alpha(T_1, \ldots, T_5) \quad \textit{for every observed sample}, \qquad \Pr_F(\mathrm{SB} \ge \mu) \ge 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 5$.

Proof. Write $p_j = p_5(j)$ and put

$$a = 1 - p_4, \quad b = p_4 - p_3, \quad c = p_3 - p_2, \quad d = p_2 - p_1, \quad e = p_1 - p_0, \quad f = p_0.$$

For ascending taints the Stringer bound is $at_1 + bt_2 + ct_3 + dt_4 + et_5 + f$. After affine normalization, it is enough to consider

$$(0, x, y, z, w, 1), \qquad 0 \le x \le y \le z \le w \le 1, \qquad s = bx + cy + dz + ew + f.$$

With $(D_0, \ldots, D_5) \sim \text{Dirichlet}(1, \ldots, 1)$, define

$$V = \Pr_D\{xD_1 + yD_2 + zD_3 + wD_4 + D_5 > s\}.$$

Put

$$\begin{aligned}
H_1 &= \frac{(1-s)^5}{(1-x)(1-y)(1-z)(1-w)}, \\
H_w &= \frac{(w-s)^5}{w(w-x)(w-y)(w-z)(1-w)}, \\
H_z &= \frac{(z-s)^5}{z(z-x)(z-y)(w-z)(1-z)}, \\
H_y &= \frac{(y-s)^5}{y(y-x)(z-y)(w-y)(1-y)}.
\end{aligned}$$

The uniform-simplex cap formula gives

$$V = \begin{cases} H_1, & s \ge w, \\ H_1 - H_w, & z \le s \le w, \\ H_1 - H_w + H_z, & y \le s \le z, \\ H_1 - H_w + H_z - H_y, & x \le s \le y, \\ 1 - s^5/(xyzw), & s \le x, \end{cases} \tag{14}$$

with boundary values defined by continuity. If $q_1 = e + f$, $q_2 = d + e + f$, $q_3 = c + d + e + f$, and $g = b + c + d + e + f$, the factor equations imply

$$\begin{aligned}
\alpha &= (1-f)^5, \\
\alpha &= (1-q_1)^4(1 + 4q_1), \\
\alpha &= (1-q_2)^3(1 + 3q_2 + 6q_2^2), \\
\alpha &= (1-q_3)^2(1 + 2q_3 + 3q_3^2 + 4q_3^3), \\
1 - \alpha &= g^5.
\end{aligned} \tag{15}$$

Let $K = (1-x)(1-y)(1-z)(1-w)$, $W = w(w-x)(w-y)(w-z)$, $Z = z(z-x)(z-y)$, and $Y = y(y-x)$. In the first four regions, multiply $\alpha - V$ by, respectively, K, KW, KWZ, and $KWZY$, and denote the cleared residuals by P_A, P_B, P_C, P_D. For the final region use $P_E = s^5 - g^5 xyzw$. Exact polynomial division after (15) gives

$$\begin{aligned}
P_B &= (1-w)R_B, \\
P_C &= (1-w)(1-z)(w-z)R_C, \\
P_D &= (1-w)(1-z)(1-y)(w-z)(w-y)(z-y)R_D.
\end{aligned}$$

The remaining polynomials P_A, R_B, R_C, R_D, P_E have degrees $5, 8, 9, 8, 5$, respectively, and every extracted factor is nonnegative.

For completeness, we describe the exact four-simplex decomposition on which their Bernstein coefficients are evaluated. Let $v_i = (0^{4-i}, 1^i)$ be the vertices of the ordered-knot simplex. If x_r denotes x, y, z, w for $r = 1, 2, 3, 4$, let $h_{r,ij}$ be the intersection of the edge v_iv_j with $s = x_r$. Successive slicing by $s = w, z, y, x$ gives the following triangulations:

$$\begin{aligned}
\mathcal{T}_A &= [v_0, h_{4,01}, h_{4,02}, h_{4,03}, h_{4,04}],\\
\mathcal{T}_B &= [h_{4,04}, h_{3,12}, h_{4,01}, h_{4,03}, h_{4,02}],\\
&\quad\ [h_{3,13}, h_{4,04}, h_{3,12}, h_{4,01}, h_{3,14}],\\
&\quad\ [h_{3,13}, h_{4,04}, h_{3,12}, h_{4,01}, h_{4,03}],\\
\mathcal{T}_C &= [h_{2,23}, h_{3,04}, h_{3,02}, h_{3,12}, h_{3,03}],\\
&\quad\ [h_{2,23}, h_{3,04}, h_{3,02}, h_{3,12}, h_{2,24}],\\
&\quad\ [h_{2,23}, h_{3,13}, h_{3,04}, h_{3,12}, h_{3,03}],\\
&\quad\ [h_{3,14}, h_{2,23}, h_{3,04}, h_{3,12}, h_{2,24}],\\
&\quad\ [h_{3,14}, h_{2,23}, h_{3,13}, h_{3,04}, h_{3,12}],\\
\mathcal{T}_D &= [h_{2,13}, h_{2,04}, h_{2,23}, h_{1,34}, h_{2,03}],\\
&\quad\ [h_{2,14}, h_{2,13}, h_{2,04}, h_{2,23}, h_{1,34}],\\
&\quad\ [h_{2,24}, h_{2,14}, h_{2,04}, h_{2,23}, h_{1,34}],\\
\mathcal{T}_E &= [h_{1,04}, h_{1,14}, h_{1,24}, h_{1,34}, v_4].
\end{aligned}$$

The listed simplices meet only on common faces and cover their respective convex regions. Their vertex formulas follow by solving one affine equation on an edge; all denominators are positive sums of the six weights.

Map each four-simplex affinely to the standard four-simplex. The respective structural-zero counts are

region	degree	number of simplices	zero counts by simplex
A	5	1	1
B	8	3	$66, 15, 39$
C	9	5	$59, 35, 92, 59, 92$
D	8	3	$15, 39, 66$
E	5	1	1.

These zeros are exact identities: the symbolic derivation proves the corresponding face-ideal powers by Gröbner reduction over $\mathbb{Q}(b, c, d, e, f)$. Each factor root is then enclosed on a 2^{-240} dyadic grid with integer-checked binomial-CDF endpoint signs. All vertex, polynomial, and Bernstein operations use integer-directed outward rounding on a 2^{-256} dyadic grid. Every coefficient not forced to zero has a strictly positive lower endpoint, and every four-simplex affine determinant is enclosed away from zero. The smallest bounds by region are shown in Table 3. The commands `make n5-structure-check` and `make n5-certificate-check` regenerate both exact layers byte-for-byte, and the numerical certificate records the SHA-256 digest of the symbolic structure. Hence every residual is nonnegative, so $V \leq \alpha$ in all five regions. Validity of G_α proves the result. $\square$

α	$s \geq w$	$z \leq s \leq w$	$y \leq s \leq z$	$x \leq s \leq y$	$s \leq x$
0.01	8.16×10^{-13}	7.45×10^{-20}	3.23×10^{-21}	7.81×10^{-16}	3.47×10^{-4}
0.05	6.70×10^{-9}	1.89×10^{-14}	6.24×10^{-16}	3.57×10^{-12}	1.38×10^{-3}
0.10	3.81×10^{-7}	4.93×10^{-12}	1.36×10^{-13}	1.34×10^{-10}	2.20×10^{-3}

Table 3: Smallest positive dyadic lower bound among the relevant Bernstein coefficients in the $n = 5$ certificate. The displayed values are decimal approximations; all endpoints and structural-zero proofs are machine-readable.

3.4 The case $n = 6$

The next dimension has six cap regions. A larger exact certificate proves all three conventional levels.

Theorem 8. *Let $n = 6$ and let $T_1, \dots, T_6$ be i.i.d. with common distribution F on $[0, 1]$ and mean μ. For $\alpha \in \{0.10, 0.05, 0.01\}$,*

$$\mathrm{SB} \geq G_\alpha(T_1, \dots, T_6) \quad \textit{for every observed sample}, \qquad \Pr_F(\mathrm{SB} \geq \mu) \geq 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 6$.

Proof. Write $p_j = p_6(j)$ and put

$$\begin{aligned} &a = 1 - p_5, \quad b = p_5 - p_4, \quad c = p_4 - p_3, \quad d = p_3 - p_2, \\ &e = p_2 - p_1, \quad f = p_1 - p_0, \quad g = p_0. \end{aligned}$$

If the smallest taint is one, all taints, Stringer, and the Gaffke limit equal one. Otherwise, subtracting the smallest taint and dividing by its distance from one gives the ascending knots and Stringer threshold

$$(t_0, \dots, t_6) = (0, x, y, z, w, u, 1), \qquad 0 \leq x \leq y \leq z \leq w \leq u \leq 1,$$
$$s = bx + cy + dz + ew + fu + g.$$

For $(D_0, \dots, D_6) \sim \text{Dirichlet}(1, \dots, 1)$, let

$$V = \Pr_D \left\{ \sum_{i=0}^{6} t_i D_i > s \right\}.$$

For distinct knots, the uniform-simplex cap formula is

$$V = \sum_{i: t_i > s} \frac{(t_i - s)^6}{\prod_{j \neq i} (t_i - t_j)}, \tag{16}$$

and coincident-knot values follow by continuity. It is enough to prove $V \leq \alpha$ at each listed value of α.

For $k = 0, \ldots, 4$, set

$$q_0 = g, \quad q_1 = f+g, \quad q_2 = e+f+g, \quad q_3 = d+e+f+g, \quad q_4 = c+d+e+f+g.$$

The Clopper–Pearson equations give

$$\alpha = \sum_{j=0}^{k} \binom{6}{j} q_k^j (1-q_k)^{6-j} \quad (k = 0, \ldots, 4), \qquad 1-\alpha = (b+c+d+e+f+g)^6. \tag{17}$$

In the first five regions of (16), clear the positive common denominator in $\alpha - V$ and substitute (17); in the last region use $s^6 - (b+c+d+e+f+g)^6 xyzwu$. Exact polynomial division extracts the following nonnegative boundary factors:

region	residual degree	extracted factors
$s \geq u$	6	1
$w \leq s \leq u$	10	$1-u$
$z \leq s \leq w$	12	$(1-w)(1-u)(u-w)$
$y \leq s \leq z$	12	$(1-z)(1-w)(1-u)(w-z)(u-z)(u-w)$
$x \leq s \leq y$	10	$\prod_{r\in\{y,z,w,u\}} (1-r) \prod_{r<r';\ r,r'\in\{y,z,w,u\}} (r'-r)$
$s \leq x$	6	$1.$

The divisions are performed in an exact polynomial ring and reject any nonzero remainder.

Let $v_i = (0^{5-i}, 1^i)$, $i = 0, \ldots, 5$, be the ordered-knot simplex vertices. Intersecting its edges with $s = u, w, z, y, x$ and slicing successively gives fixed triangulations containing respectively $1, 5, 10, 10, 5, 1$ five-simplices. Every intersection coordinate is retained as an exact rational function of $b, \ldots, g$. Exact boundary identities and directed strict-sign checks place every vertex in its claimed convex region, and all 210 pairs of the 21 named vertices are certified as distinct. The oriented-chain check pairs all 48 internal four-facets with opposite orientations. The 96 unpaired four-facets lie on the six outer ordered-simplex hyperplanes, 16 on each, and the directed determinant sum lies strictly between $1/2$ and $3/2$. Thus the chain represents an integer multiple of the generator of $H_5(\Delta, \partial\Delta; \mathbb{Z})$, and its determinant sum is that integer times the normalized volume of Δ, which is one. The integer is therefore one. Because every simplex has positive orientation and lies in Δ, the 32 simplices cover the domain without interior multiplicity. After mapping each five-simplex to the standard five-simplex, the six residual degrees, simplex counts, and total structural-zero counts are

region	degree	five-simplices	structural zeros
A	6	1	1
B	10	5	1433
C	12	10	7514
D	12	10	7514
E	10	5	1433
F	6	1	1.

The structural zeros are exact. For each relevant affine face ideal I, the symbolic layer verifies the required membership in I^q by reducing the residual and every ordinary partial derivative of total order less than q modulo I. These are exact Singular calculations over $\mathbb{Q}(b,c,d,e,f,g)$; the weights are not specialized. A rational specialization is used only to select independent generators from redundant face equations. At the certified weights, directed intervals enclose a full-rank normal minor away from zero for every selected generator set, so the generic face identities specialize validly. For the remaining coefficients, each factor root is bracketed on a 2^{-320} dyadic grid with integer-checked binomial-CDF endpoint signs. All subsequent vertex, determinant, polynomial, and Bernstein operations use integer-directed outward rounding on a 2^{-384} grid. Every five-simplex determinant is enclosed away from zero, and every nonstructural coefficient has a strictly positive lower endpoint. The commands `make n6-structure-check` and `make n6-certificate-check` regenerate the two machine-readable layers, whose link is fixed by a SHA-256 digest. Hence every cleared residual is nonnegative and $V \le \alpha$ at all three listed levels. Validity of G_α proves the coverage statement. $\square$

α	A	B	C	D	E	F
0.10	1.80×10^{-8}	1.85×10^{-15}	1.93×10^{-19}	4.24×10^{-19}	1.35×10^{-13}	1.53×10^{-3}
0.05	1.23×10^{-10}	1.17×10^{-18}	6.57×10^{-23}	4.01×10^{-22}	1.66×10^{-15}	9.57×10^{-4}
0.01	2.02×10^{-15}	9.63×10^{-26}	1.22×10^{-30}	6.05×10^{-29}	6.29×10^{-20}	2.41×10^{-4}

Table 4: Smallest positive dyadic lower bound by cap region in the $n = 6$ certificate. The displayed values are decimal approximations; every interval endpoint and structural-zero proof is machine-readable.

4 Poisson factors at conventional audit confidence levels

For $0 \le j \le n$, let $p_n^{\mathrm{B}}(j)$ denote the binomial factor used above, and let

$$p_n^{\mathrm{P}}(j) = \frac{\lambda_j}{n}, \qquad \sum_{i=0}^{j} e^{-\lambda_j} \frac{\lambda_j^i}{i!} = \alpha,$$

denote its Poisson analogue. The Poisson factors are not truncated at one; as in audit factor tables, a factor may exceed one.

Proposition 9 (All-n factor domination)**.** *If $0 < \alpha < e^{-1}$, then*

$$p_n^{\mathrm{P}}(j) \ge p_n^{\mathrm{B}}(j), \qquad j = 0, \dots, n, \quad n \ge 1.$$

Consequently, the Poisson-factor Stringer bound is at least the binomial-factor Stringer bound for every observed sample.

Proof. Write $B(j; n, p) = \Pr\{\mathrm{Bin}(n,p) \le j\}$ and $P(j;\lambda) = \Pr\{\mathrm{Pois}(\lambda) \le j\}$. Anderson and Samuels [3, Corollary 2.1 and Section 4.2] show that

$$B(j; n, \lambda/n) < P(j; \lambda) \quad \text{when } j < \lambda - 1 \text{ and } \lambda < n.$$

Because $P(j;\lambda_j) = \alpha < e^{-1} = P(0;1)$, their Theorem 4.1 gives $\lambda_j > j+1$. If $\lambda_j < n$, the displayed inequality yields $B(j;n,p_n^{\mathrm{P}}(j)) < \alpha$; since $B(j;n,p)$ is decreasing in p, its root satisfies $p_n^{\mathrm{B}}(j) < p_n^{\mathrm{P}}(j)$. If $\lambda_j \geq n$, the conclusion follows instead from $p_n^{\mathrm{P}}(j) \geq 1 \geq p_n^{\mathrm{B}}(j)$.

For the pointwise comparison, put $d_j = p_n^{\mathrm{P}}(j) - p_n^{\mathrm{B}}(j) \geq 0$. If $t_{(1)} \geq \cdots \geq t_{(n)}$, summation by parts gives

$$\mathrm{SB_P} - \mathrm{SB_B} = d_0(1 - t_{(1)}) + \sum_{j=1}^{n-1} d_j(t_{(j)} - t_{(j+1)}) + d_n t_{(n)} \geq 0.$$

□

Corollary 10. *The Poisson-factor Stringer bound is distribution-free conservative at $n = 2$ whenever $0 < \alpha < e^{-1}$, and at $n = 3$, $n = 4$, $n = 5$, and $n = 6$ for $\alpha \in \{0.10, 0.05, 0.01\}$.*

Proof. Combine Proposition 9 with Theorems 1, 5, 6, 7, and 8. □

The factor comparison is not the only route to a Poisson guarantee. The following lemma uses a simultaneous survival band with the index required at the right side of each sample point.

Lemma 11 (Simultaneous-band lower bound)**.** *Let $T_1, \ldots, T_n$ be i.i.d. with common distribution F on $[0,1]$ and mean μ, and let $t_{(1)} \geq \cdots \geq t_{(n)}$ be the observed order statistics. Let $0 \leq c_0 \leq \cdots \leq c_n$ with $c_n \geq 1$, and form*

$$L_c = c_0 + \sum_{j=1}^{n}(c_j - c_{j-1})t_{(j)}.$$

Put $a_i = \min\{1, c_{i-1}\}$ and let

$$Q_n(a_1, \ldots, a_n) = \Pr\{V_{i:n} \leq a_i,\ i = 1, \ldots, n\},$$

where $V_{1:n} \leq \cdots \leq V_{n:n}$ are uniform order statistics. Then

$$\Pr_F\{L_c \geq \mu\} \geq Q_n(a_1, \ldots, a_n).$$

If $\bar{c}_j = \min\{1, c_j\}$ and $L_{\bar{c}}$ is formed from those factorwise-capped coefficients, then the same lower bound holds and $L_{\bar{c}} \leq \min\{1, L_c\}$ pointwise.

Proof. Write $G(x) = \Pr_F\{T > x\}$ and $N_x = \#\{r : T_r > x\}$. Apart from the finitely many sampled values, which do not affect the integral,

$$L_c = \int_0^1 c_{N_x}\, dx, \qquad \mu = \int_0^1 G(x)\, dx. \tag{18}$$

To include atomic F, augment the experiment with independent $R_r \sim \mathrm{Unif}(0,1)$ and define

$$V_r = \Pr_F\{T > T_r\} + R_r \Pr_F\{T = T_r\}.$$

The randomized probability integral transform makes the V_r i.i.d. uniform. If $N_x = j < n$, the j observations above x have $V_r \leq G(x)$ and the remaining observations have $V_r \geq G(x)$.

Hence $G(x) \leq V_{j+1:n}$. On the event in the definition of Q_n, this gives $G(x) \leq a_{j+1} \leq c_j$. If $N_x = n$, use $G(x) \leq 1 \leq c_n$. Integration in (18) proves the event inclusion and hence the result. The coverage event depends only on the original sample, so its probability in the augmented experiment is its ordinary sampling probability. The capped factors are nondecreasing, end at one, and induce the same boundaries a_i. The first conclusion therefore applies to them. Finally, summation by parts gives

$$L_c = c_0(1 - t_{(1)}) + \sum_{j=1}^{n-1} c_j(t_{(j)} - t_{(j+1)}) + c_n t_{(n)}.$$

All displayed weights are nonnegative and sum to one. Since $\bar{c}_j \leq c_j$ and $\bar{c}_n = 1$, this representation gives $L_{\bar{c}} \leq L_c$ and $L_{\bar{c}} \leq 1$. □

For nondecreasing boundaries $a_1, \ldots, a_n$, Bolshev's recursion [19, pp. 366–367] evaluates the probability in the lemma exactly. With $Q_0 = 1$, it is

$$Q_m = 1 - \sum_{i=1}^{m} \binom{m}{i} (1 - a_{m-i+1})^i Q_{m-i}, \qquad m = 1, \ldots, n. \tag{19}$$

Theorem 12 (Direct Poisson guarantees at larger sample sizes)**.** *Let the observations be i.i.d. with an arbitrary common distribution F on $[0,1]$. The Poisson-factor Stringer bound is conservative throughout the following ranges:*

nominal confidence	*every certified sample size*
90%	$1 \leq n \leq 8$
95%	$1 \leq n \leq 11$
99%	$1 \leq n \leq 20.$

The conclusion also holds if the Poisson factors are capped at one before forming the bound.

Proof. Use Lemma 11 with $c_j = p_n^{\mathrm{P}}(j) = \lambda_j/n$; these factors are increasing in j. At the listed levels, $\alpha < e^{-1}$, so $\lambda_n > n + 1$ by the argument in Proposition 9; hence $c_n > 1$.

For each required j, the certificate encloses λ_j between adjacent dyadic rationals $L_j < U_j$ on the grid $2^{-80}\mathbb{Z}$. The endpoint signs

$$e^{-L_j} \sum_{k=0}^{j} \frac{L_j^k}{k!} > \alpha, \qquad e^{-U_j} \sum_{k=0}^{j} \frac{U_j^k}{k!} < \alpha$$

are checked with exact rational bounds for the exponential. Thus $L_j < \lambda_j < U_j$. Substitution of the smaller boundaries $a_i^- = \min\{1, L_{i-1}/n\}$ into (19) gives an exact rational lower bound for the event probability in Lemma 11. The certificate checks every positive n in the three stated ranges. At the largest n its lower bounds are as follows; the displayed decimals summarize exact rationals.

α	largest n	$Q_n(a_1^-, \ldots, a_n^-)$	margin over $1 - \alpha$
0.10	8	0.9028501343687063	0.0028501343687063
0.05	11	0.9532937128380234	0.0032937128380234
0.01	20	0.9906297230587641	0.0006297230587641

Lemma 11 completes the proof. Replacing each c_j by $\min(1, c_j)$ preserves the same pointwise-band argument. $\square$

All arithmetic in this certificate is exact. After power-of-two range reduction to $[0, 1]$, consecutive odd and even partial sums of the alternating exponential series enclose e^{-x}; the code then evaluates (19) with rational arithmetic. At the next sample size, even the upper endpoint brackets give simultaneous-event probabilities 0.8903656999, 0.9482646529, and 0.9899786984, respectively, each below nominal. This limits the sufficient-event proof, not the Stringer bound: the integral inequality can hold when the pointwise band crosses G.

The same sufficient event gives an all-sample-size calibration while retaining the familiar Poisson factor curve. For $0 < \alpha < 1$, define λ_j as above and put

$$\begin{aligned}
\kappa_0 &= \max\left\{1, \frac{n}{\lambda_n}\right\}, \\
a_i(\kappa) &= \min\left\{1, \frac{\kappa\lambda_{i-1}}{n}\right\}, \qquad i = 1, \dots, n, \\
Q_{n,\alpha}(\kappa) &= Q_n\big(a_1(\kappa), \dots, a_n(\kappa)\big), \\
\kappa_{n,\alpha} &= \inf\{\kappa \geq \kappa_0 : Q_{n,\alpha}(\kappa) \geq 1 - \alpha\}.
\end{aligned} \tag{20}$$

The function $Q_{n,\alpha}$ is continuous and nondecreasing; its superlevel set in (20) is therefore closed.

Theorem 13 (All-sample-size scalar Poisson calibration)**.** *Let the observations be i.i.d. with an arbitrary common distribution on* $[0, 1]$*, and let* $0 < \alpha < 1$*. Then*

$$U_{n,\alpha} = \min\{1, \kappa_{n,\alpha}\mathrm{SB}_\mathrm{P}\}$$

has coverage at least $1 - \alpha$ *for every* $n \geq 1$*. The same conclusion holds with any certified* $\kappa \geq \kappa_{n,\alpha}$*. It also holds if each Poisson factor is capped at one before the complete factor-capped Stringer expression is multiplied by* κ*. A pointwise no-larger valid rule uses the calibrated factors*

$$\widehat{c}_j = \min\left\{1, \frac{\kappa\lambda_j}{n}\right\}$$

directly in the Stringer formula; the same curve results if the ordinary factors are capped before calibration.

Proof. Multiplying every Poisson factor by κ multiplies the complete Stringer expression by κ. Thus use Lemma 11 with $c_j = \kappa\lambda_j/n$. These factors are nondecreasing, and $c_n \geq 1$ for every $\kappa \geq \kappa_0$. The lemma and (20) give

$$\mathbb{P}\{\kappa\mathrm{SB}_\mathrm{P} \geq \mu\} \geq Q_{n,\alpha}(\kappa).$$

The set in (20) is nonempty: because the λ_j are increasing, $\kappa = \max\{1, n/\lambda_0\}$ belongs to the permitted domain, makes every $a_i(\kappa)$ equal to one, and hence makes $Q_{n,\alpha}(\kappa) = 1$. Continuity shows that the infimum is attained. Use $\kappa = \kappa_{n,\alpha}$ in the preceding inequality. Finally, capping at one preserves coverage because $\mu \leq 1$.

For the factor-capped convention, put $\bar{p}_j = \min\{1, \lambda_j/n\}$ and use $c_j = \kappa\bar{p}_j$. Because $\kappa \geq 1$,

$$\min\{1, c_{i-1}\} = \min\left\{1, \frac{\kappa\lambda_{i-1}}{n}\right\} = a_i(\kappa),$$

and the definition of κ_0 gives $c_n \geq 1$. Thus the same event and the same argument apply. Finally, apply the factorwise-capping conclusion of Lemma 11. Because $\kappa \geq 1$, $\min\{1, \kappa\bar{p}_j\} = \widehat{c}_j$, so both base-factor conventions give the stated common curve, pointwise no larger than their final-cap reports. □

The exact joint calibration in (20) is readily precomputed, but the next result gives a closed-form upper bound that does not depend on the sample size.

Corollary 14 (Sample-size-uniform multiplier)**.** *Suppose $0 < \alpha \leq e^{-1}$ and $\kappa > 1$ satisfies*

$$\alpha^{\kappa-1} \leq 1 - \kappa e^{1-\kappa}. \tag{21}$$

Then $\kappa_{n,\alpha} \leq \kappa$ for every $n \geq 1$. Consequently, both the final-cap and calibrated-factor-capped full-scale rules formed with this κ have coverage at least $1-\alpha$ at every sample size. In particular,

$$\kappa_{n,\alpha} \leq 2 \qquad \textit{whenever} \qquad \alpha \leq 1 - \frac{2}{e}.$$

Thus the multiplier two is a distribution-free valid choice, uniformly in n, at every nominal confidence level at least $2/e \approx 73.5759\%$.

Proof. Let $\lambda_0 = -\log\alpha$ and let $X_a \sim \mathrm{Gamma}(a, 1)$. Because $\alpha \leq e^{-1}$, $c = \lambda_0 - 1 \geq 0$. The gamma-tail monotonicity theorem of Pinelis [17] states that $\Pr\{X_a - a > c\}$ is increasing in a. Hence, for every integer $j \geq 0$,

$$\Pr\{X_{j+1} > j + \lambda_0\} \geq \Pr\{X_1 > \lambda_0\} = \alpha, \qquad \text{so} \qquad \lambda_j \geq j + \lambda_0. \tag{22}$$

Here we used the Erlang–Poisson identity $\Pr\{X_{j+1} > x\} = P(j; x)$ and the definition $P(j; \lambda_j) = \alpha$. In particular, the terminal condition in Lemma 11 holds for every $\kappa > 1$.

Put $\rho = \kappa e^{1-\kappa} < 1$. If $a_{j+1}(\kappa) = \kappa\lambda_j/n < 1$, the Anderson–Samuels comparison used in Proposition 9 gives

$$\begin{aligned}
\Pr\{V_{j+1:n} > a_{j+1}(\kappa)\} &= B\left(j; n, \frac{\kappa\lambda_j}{n}\right) \\
&< P(j; \kappa\lambda_j) \\
&\leq \alpha\,\kappa^j e^{-(\kappa-1)\lambda_j} \leq \alpha^\kappa \rho^j.
\end{aligned}$$

The comparison applies because (22), $\lambda_0 \geq 1$, and $\kappa > 1$ imply $j < \kappa\lambda_j - 1$. The first weak inequality in the last two lines uses $\sum_{r=0}^{j} \kappa^r \lambda_j^r / r! \leq \kappa^j \sum_{r=0}^{j} \lambda_j^r / r!$; the last uses (22). If the boundary is one, the probability on the left is zero. A union bound therefore yields

$$1 - Q_{n,\alpha}(\kappa) \leq \alpha^\kappa \sum_{j=0}^{n-1} \rho^j \leq \frac{\alpha^\kappa}{1-\rho} \leq \alpha$$

by (21). Thus κ belongs to the defining set in (20). The coverage statements follow from Theorem 13. For $\kappa = 2$, (21) reduces to $\alpha \leq 1 - 2/e$. □

The geometric bound need not be used on the first few crossing events. Retaining their exact Poisson probabilities gives a sharper criterion.

Corollary 15 (Finite-prefix uniform multiplier)**.** *Under the assumptions of Corollary 14, fix an integer $J \geq 1$. If*

$$\sum_{j=0}^{J-1} P(j;\kappa\lambda_j) + \frac{\alpha^\kappa \rho^J}{1-\rho} \leq \alpha, \qquad \rho = \kappa e^{1-\kappa}, \tag{23}$$

then $\kappa_{n,\alpha} \leq \kappa$ for every $n \geq 1$.

Proof. In the union bound from the preceding proof, retain $P(j;\kappa\lambda_j)$ for $j < J$. For $j \geq J$, the same calculation gives $P(j;\kappa\lambda_j) \leq \alpha^\kappa \rho^j$. For $n < J$, retain only the available terms and then enlarge the bound by adding the missing nonnegative terms through $J-1$. Thus, for every n,

$$1 - Q_{n,\alpha}(\kappa) \leq \sum_{j=0}^{J-1} P(j;\kappa\lambda_j) + \alpha^\kappa \sum_{j=J}^{\infty} \rho^j \leq \alpha.$$

The conclusion follows as in Corollary 14. □

The following table compares the refined choices with the elementary closed-form choices from (21). Exact rational exponential enclosures certify both columns; the refined column uses $J = 100$.

nominal confidence	100-term certified κ	closed-form κ
90%	1.53	1.76
95%	1.44	1.66
99%	1.33	1.52

Table 5: Simple full-scale Poisson calibration multipliers valid for every sample size. These are sufficient choices, not optimality claims.

The full-scale rule raises the no-error factor. An anchored calibration can leave that factor unchanged. Put $\bar{p}_j = \min\{1, \lambda_j/n\}$. If $\bar{p}_0 < 1$, define

$$\begin{aligned} \eta_0 &= \max\left\{1, \frac{1-\bar{p}_0}{\bar{p}_n - \bar{p}_0}\right\}, \\ d_j(\eta) &= \bar{p}_0 + \eta(\bar{p}_j - \bar{p}_0), \\ b_i(\eta) &= \min\{1, d_{i-1}(\eta)\}, \\ R_{n,\alpha}(\eta) &= Q_n\big(b_1(\eta), \ldots, b_n(\eta)\big), \\ \eta_{n,\alpha} &= \inf\{\eta \geq \eta_0 : R_{n,\alpha}(\eta) \geq 1-\alpha\}. \end{aligned} \tag{24}$$

If $\bar{p}_0 = 1$, set $\eta_{n,\alpha} = 1$.

Corollary 16 (Zero-taint-preserving Poisson calibration)**.** *Let* $\mathrm{SB}_{\bar{\mathrm{P}}}$ *use the factor-capped curve $\bar{p}_j$, and let $p_j^{\mathrm{P}} = \lambda_j/n$ denote the untruncated factors. Both*

$$\begin{aligned} U_{n,\alpha}^{\mathrm{A}} &= \min\{1, \bar{p}_0 + \eta_{n,\alpha}(\mathrm{SB}_{\bar{\mathrm{P}}} - \bar{p}_0)\}, \\ U_{n,\alpha}^{\mathrm{A},u} &= \min\{1, p_0^{\mathrm{P}} + \eta_{n,\alpha}(\mathrm{SB}_{\mathrm{P}} - p_0^{\mathrm{P}})\} \end{aligned}$$

have coverage at least $1-\alpha$ for every $n \geq 1$ and every $0 < \alpha < 1$. Moreover, $\eta_{n,\alpha} \geq \kappa_{n,\alpha}$. On an all-zero sample, each returns its ordinary zero-taint factor before the final cap. The same coverage holds after capping every effective anchored factor at one; the capped and untruncated base curves then give the same factor sequence and a report pointwise no larger than either displayed rule.

Proof. If $\bar{p}_0 = 1$, monotonicity gives $\bar{p}_j = 1$ for every j, while $p_0^{\mathrm{P}} \geq 1$. Moreover, $\kappa_0 = 1$ and $Q_{n,\alpha}(1) = 1$, so $\kappa_{n,\alpha} = \eta_{n,\alpha} = 1$. Both displayed rules equal one after the final cap. Hence assume $\bar{p}_0 < 1$. The factors $d_j(\eta)$ are nondecreasing and $d_n(\eta) \geq 1$ on the permitted domain, so Lemma 11 gives coverage at least $R_{n,\alpha}(\eta)$. The feasible set in (24) is nonempty. Indeed, every boundary except the first tends to one as $\eta \to \infty$, whereas $b_1(\eta) = \bar{p}_0$. In the nontrivial case, $\bar{p}_0 = \lambda_0/n < 1$ and $\lambda_0 = -\log \alpha$, so

$$\lim_{\eta \to \infty} R_{n,\alpha}(\eta) = 1 - (1 - \lambda_0/n)^n > 1 - e^{-\lambda_0} = 1 - \alpha.$$

Continuity shows that the infimum is finite and attained. Linearity of the Stringer formula gives the first displayed rule.

For the untruncated curve, use $d_j^u = p_0^{\mathrm{P}} + \eta(p_j^{\mathrm{P}} - p_0^{\mathrm{P}})$. If $p_j^{\mathrm{P}} < 1$, it has the same capped boundary as d_j; if $p_j^{\mathrm{P}} \geq 1$, both boundaries equal one because $\eta \geq 1$. The terminal condition also holds. Thus the same event proves the second rule. Capping either effective factor curve at one preserves the event and gives identical factors, so the final assertion follows from Lemma 11. If $\bar{p}_n < 1$, then

$$\eta_0 = \frac{1 - \bar{p}_0}{\bar{p}_n - \bar{p}_0} \geq \frac{1}{\bar{p}_n} = \kappa_0;$$

if $\bar{p}_n = 1$, both domain lower bounds equal one. For every $x \geq \eta_0$,

$$b_i(x) \leq \min\{1, x\bar{p}_{i-1}\} = a_i(x),$$

so $R_{n,\alpha}(x) \leq Q_{n,\alpha}(x)$. Minimality therefore gives $\eta_{n,\alpha} \geq \kappa_{n,\alpha}$. If every taint is zero, all factor-increment terms vanish. $\square$

The anchored multiplier is never smaller, but it applies only to the error increments. The calibration path must be pre-specified; taking the smaller of the two bounds after observing the sample is not justified by these results. At 95% confidence, simple six-decimal valid anchored multipliers at $n = 25, 50, 100$, and 200 are 1.511563, 1.947538, 2.367560, and 2.778290, respectively; exact dyadic brackets are recorded with the full-scale certificates below.

For the full-scale path, an explicit analytic upper bound is available at the conventional levels considered here. Put $\theta = \alpha/n$, let $\lambda_j(\theta)$ solve

$$e^{-\lambda_j(\theta)} \sum_{r=0}^{j} \frac{\lambda_j(\theta)^r}{r!} = \theta,$$

and define

$$\kappa_{\mathrm{B}} = \max\left\{\kappa_0, \max_{0 \leq j < n} \frac{\lambda_j(\theta)}{\lambda_j}\right\}.$$

For $b_i = \min\{1, \lambda_{i-1}(\theta)/n\}$, the Anderson–Samuels comparison used in Proposition 9 gives

$$\mathbb{P}(V_{i:n} > b_i) = \mathbb{P}\{\mathrm{Bin}(n, b_i) \leq i - 1\} \leq \theta.$$

If $\lambda_{i-1}(\theta) \geq n$, the probability is zero; otherwise the comparison applies directly. The union bound gives $\mathbb{P}(V_{i:n} \leq b_i \ \forall i) \geq 1 - n\theta = 1 - \alpha$. Since $a_i(\kappa_{\mathrm{B}}) \geq b_i$,

$$\kappa_{n,\alpha} \leq \kappa_{\mathrm{B}} < \infty. \tag{25}$$

The exact joint calculation yields a scalar no larger than the union-bound construction in (25). Bolshev's recursion and opposite endpoints of exact dyadic brackets for the λ_j certify the smallest scalar under this sufficient-event criterion to within 2^{-28}. The following entries are simple six-decimal valid choices obtained by rounding the certified upper endpoints upward.

nominal confidence	$n = 25$	$n = 50$	$n = 100$	$n = 200$
90%	1.189306	1.251208	1.286176	1.305049
95%	1.126246	1.195804	1.235956	1.257979
99%	1.027950	1.111273	1.161122	1.189239

Table 6: Certified scalar multipliers for the full-scale all-sample-size band-calibrated Poisson rule. Each displayed value is an upward rounding of an exact dyadic upper endpoint.

These values multiply the complete ordinary Poisson result. They are not evidence that ordinary Stringer undercovers, nor are they claimed to be the smallest corrections among all valid procedures; they are the smallest scalar multipliers certified by (20), at the stated dyadic resolution. This construction is related in motivation to Bimpeh's exploration of adjusted significance levels [6, Section 5.4], but it uses the corrected order-statistic event and its exact joint probability. In an implementation, the underlying Poisson factors must also be exact or conservatively rounded; upward rounding of the scalar alone does not control factor-table rounding.

Thus the factor comparison previously checked numerically at 90% and 95% is analytic and holds at every sample size. In particular, Theorem 1 implies finite-sample conservatism of the Poisson-factor Stringer bound at $n = 2$ throughout this confidence range. Theorems 5, 6, 7, and 8 give the same conclusion from $n = 3$ through $n = 6$ for the three certified conventional levels. Theorem 12 extends the proved Poisson ranges to $n = 8$, $n = 11$, and $n = 20$ at 90%, 95%, and 99%, respectively. Theorem 13 supplies a conservative scalar-calibrated rule beyond those ranges without claiming coverage for the ordinary Poisson bound. Corollary 14 supplies closed-form full-scale multipliers that are valid simultaneously for every sample size. Corollary 15 sharpens them by retaining a finite prefix of the crossing probabilities. More generally, any distributional class for which the binomial-factor bound is proved conservative inherits the same conclusion for the Poisson-factor bound. Proposition 9 does not, by itself, settle coverage in the remaining cases, because the binomial-factor conjecture remains unresolved there.

5 An all-sample-size reporting safeguard

The valid comparison bound also gives a directly computable fallback outside the sample sizes for which ordinary Stringer has been proved. For $M \in \{\mathrm{B}, \mathrm{P}\}$, define

$$\mathrm{SB}^{\mathrm{safe}}_{M,\alpha}(t_1, \ldots, t_n) = \max\{\mathrm{SB}_{M,\alpha}(t_1, \ldots, t_n), G_\alpha(t_1, \ldots, t_n)\}.$$

Proposition 17 (Stringer–Gaffke safeguard). *Let $T_1, \dots, T_n$ be independent $[0,1]$-valued observations with a common mean μ. For every $n \geq 1$, every $\alpha \in (0,1)$, and either factor convention M,*

$$\Pr\{\mathrm{SB}^{\mathrm{safe}}_{M,\alpha} \geq \mu\} \geq 1 - \alpha.$$

For $n \in \{3,4,5,6\}$ and $\alpha \in \{0.10, 0.05, 0.01\}$, the safeguarded value equals the ordinary Stringer value pointwise for both factor conventions.

Proof. Finite-sample validity of the Gaffke limit gives $\Pr\{G_\alpha(T_1, \dots, T_n) \geq \mu\} \geq 1 - \alpha$. Since the maximum is pointwise at least G_α, the first conclusion follows. At the listed sample sizes and levels, Theorems 5, 6, 7, and 8 give $\mathrm{SB}_{\mathrm{B},\alpha} \geq G_\alpha$ on every sample. Proposition 9 gives $\mathrm{SB}_{\mathrm{P},\alpha} \geq \mathrm{SB}_{\mathrm{B},\alpha}$, proving the second conclusion. □

The proposition validates the pre-specified maximum, not ordinary Stringer in cases where Stringer is smaller. The repository implementation evaluates the Dirichlet-average tail at rational points by exact confluent divided differences and returns an exact-sign dyadic upper bracket for G_α. Thus the operational floor does not depend on trusting a floating-point B-spline calculation.

5.1 An all-sample-size one-cap region

The safeguard is known to be identical to Stringer on every sample at $n = 3$, $n = 4$, $n = 5$, and $n = 6$ at the three certified levels. The following result identifies a directly checkable identity region without a sample-size cutoff.

Proposition 18 (One-cap comparison). *Let $n \geq 1$ and $0 < \alpha \leq 1/4$. Arrange the observed taints as $x_0 \leq \cdots \leq x_{n-1}$, and let s_{B} be their binomial-factor Stringer value. If $s_{\mathrm{B}} \geq x_{n-1}$, then*

$$s_{\mathrm{B}} \geq G_\alpha(x_0, \dots, x_{n-1}).$$

Consequently, on that sample the pre-specified safeguard equals ordinary Stringer for both the binomial and Poisson factor conventions.

Proof. We first give an analytic cap bound with no restriction on n. Let $x_n = 1$, let $c_i > 0$ with $\sum_{i=0}^{n} c_i = 1$, and put

$$s = \sum_{i=0}^{n} c_i x_i, \qquad C_r = \sum_{i=0}^{r-1} c_i \quad (1 \leq r \leq n).$$

For $D \sim \mathrm{Dirichlet}(1, \dots, 1)$, if $s \geq x_{n-1}$, the one-upper-knot cap formula gives

$$V := \Pr_D\left\{\sum_{i=0}^{n} x_i D_i > s\right\} = \frac{(1-s)^n}{\prod_{i=0}^{n-1}(1 - x_i)}. \tag{26}$$

Degenerate boundary cases follow directly or by continuity. Otherwise set $S = 1 - s$, $y_i = 1 - x_i$, and $u_i = y_i/S - 1$. Then

$$u_0 \geq \cdots \geq u_{n-1} \geq 0, \qquad \sum_{i=0}^{n-1} c_i u_i = c_n, \qquad V = \prod_{i=0}^{n-1}(1 + u_i)^{-1}.$$

Writing $u_n = 0$ and letting v_r denote the vector with its first r entries equal to one and the rest zero, define

$$\lambda_r = \frac{C_r(u_{r-1} - u_r)}{c_n}.$$

The λ_r are nonnegative, sum to one, and

$$u = \sum_{r=1}^{n} \lambda_r \frac{c_n}{C_r} v_r.$$

Since $\sum_i \log(1 + u_i)$ is concave, its value on this convex hull is at least its smallest vertex value. Hence

$$V \le \max_{1 \le r \le n} \left(\frac{C_r}{C_r + c_n} \right)^r. \tag{27}$$

For the ascending-knot binomial Stringer representation,

$$c_0 = 1 - p_n(n-1), \quad c_j = p_n(n-j) - p_n(n-j-1) \ (1 \le j < n), \quad c_n = p_n(0),$$

so $C_r = 1 - p_n(n-r)$. The $r = n$ term in (27) equals $(1 - p_n(0))^n = \alpha$. For every $r < n$, every n, and $0 < \alpha \le 1/4$, we now prove

$$\left(\frac{1 - p_n(n-r)}{1 - p_n(n-r) + p_n(0)} \right)^r \le \alpha. \tag{28}$$

Write $\alpha = e^{-x}$, $a = e^{-x/n}$, and $b = e^{-x/r}$, and define

$$q_{n,r} = \frac{b(1-a)}{1-b} = \frac{1 - e^{-x/n}}{e^{x/r} - 1}. \tag{29}$$

By binomial symmetry, $q = 1 - p_n(n-r)$ is the unique solution of $\Pr\{\mathrm{Bin}(n,q) \ge r\} = \alpha$. Inequality (28) is equivalent to $q \le q_{n,r}$. It therefore suffices to show

$$\Pr\{\mathrm{Bin}(n, q_{n,r}) \ge r\} \ge \alpha. \tag{30}$$

First suppose $r \ge 2$, and put $\mu_n = nq_{n,r}$. For fixed r, μ_n increases with n, because

$$\frac{d}{dz}\{z(1 - e^{-x/z})\} = 1 - (1 + x/z)e^{-x/z} > 0.$$

Also $q_{n,r} < r/n$, because the numerator in (29) is less than x/n and the denominator is greater than x/r. If $\mu_n \ge r - 1$, then a binomial variable X with this mean satisfies $\{X \ge r\} = \{X > \mathbb{E}X\}$. Since $nq_{n,r} \ge 1 > \log(4/3)$, the sharp binomial-mean inequality of Pinelis [16] gives

$$\Pr(X \ge r) \ge \frac{1}{4} \ge \alpha.$$

If instead $\mu_n < r - 1$, let $\mu_0 = \mu_{r+1}$. Monotonicity in the success probability, followed by the fixed-mean binomial monotonicity of Anderson and Samuels [3, Theorem 2.1], gives

$$\begin{aligned} \Pr\{\mathrm{Bin}(n, q_{n,r}) \ge r\} &\ge \Pr\{\mathrm{Bin}(n, \mu_0/n) \ge r\} \\ &\ge \Pr\{\mathrm{Bin}(r+1, q_{r+1,r}) \ge r\}, \end{aligned} \tag{31}$$

because $r-1>\mu_0$.

It remains to bound the last probability. Set

$$u=\frac{x}{r+1},\qquad v=\frac{x}{r},\qquad A=\frac{1-e^{-u}}{1-e^{-v}},$$

so $q_{r+1,r}=bA$ and $\alpha=b^r$. The two-term boundary tail is

$$\Pr\{\mathrm{Bin}(r+1,bA)\ge r\}=\alpha A^r(r+1-rbA). \tag{32}$$

Write

$$M_r(x)=A^r(r+1-rbA),$$

the multiplier of α in (32). This multiplier is strictly increasing in x. Indeed, with $\psi(z)=z/(e^z-1)$,

$$\frac{d}{dx}\log A=\frac{\psi(u)-\psi(v)}{x}>0.$$

Moreover, ψ is decreasing and

$$\psi'(z)+1=\frac{e^z(e^z-1-z)}{(e^z-1)^2}>0.$$

Consequently,

$$\frac{d}{dx}\log(bA)=-\frac{1}{r}+\frac{\psi(u)-\psi(v)}{x}<-\frac{1}{r}+\frac{v-u}{x}=-\frac{1}{r+1}<0.$$

Thus both factors defining $M_r(x)$ increase with x.

It remains to prove $M_r(1)>1$. Put $t=1/r$ and $d=t^2/(1+t)$. At $x=1$,

$$1-A=\frac{e^d-1}{e^t-1}=:\delta.$$

For $0<z<2$, the elementary bounds

$$z+\frac{z^2}{2}<e^z-1<\frac{z}{1-z/2},\qquad 1-z<e^{-z}<1-z+\frac{z^2}{2}$$

follow from the exponential series and $2\,\mathrm{arctanh}(z/2)>z$. They give

$$\delta<\frac{d/(1-d/2)}{t+t^2/2},\qquad \delta>\frac{d(1-t/2)}{t}.$$

Since $(e^z-1)/z$ is increasing, we also have $\delta<d/t=1/(r+1)$ and hence $A>r/(r+1)$. The second bound, together with $b=e^{-t}>1-t$ and $1-b>t-t^2/2$, gives

$$\begin{aligned} r+1-rbA&=1+r(1-b)+rb\delta\\ &>1+\left(1-\frac{t}{2}\right)+\frac{(1-t)(1-t/2)}{1+t}=\frac{3}{1+t}=\frac{3r}{r+1}. \end{aligned}$$

Therefore, for $r \geq 5$,

$$M_r(1) > 3\left(\frac{r}{r+1}\right)^{r+1} \geq 3\left(\frac{5}{6}\right)^6 > 1,$$

because the sequence $(r/(r+1))^{r+1}$ is increasing. For the remaining cases, the first bound on δ gives the following exact rational lower margins:

r	A is greater than	$\frac{3r}{r+1}A^r - 1$ is greater than
2	39/55	17/3025
3	125/161	885001/16693124
4	287/351	1841131309/25297477335.

Thus $M_r(1) > 1$ for every $r \geq 2$. Because $x \geq \log 4 > 1$ throughout the stated range, the boundary probability in (31) is at least α.

For $r = 1$, direct calculation completes the argument. In this case

$$q_{n,1} = \frac{\alpha}{1-\alpha}(1-\alpha^{1/n}), \qquad nq_{n,1} \geq \frac{2\alpha}{1+\sqrt{\alpha}} \geq \frac{\alpha}{1-\alpha} \geq -\log(1-\alpha),$$

where the middle inequality holds for $\alpha \leq 1/4$, as it does throughout the stated range. Therefore

$$\Pr\{\mathrm{Bin}(n, q_{n,1}) \geq 1\} = 1 - (1-q_{n,1})^n \geq 1 - e^{-nq_{n,1}} \geq \alpha.$$

This proves (30), hence (28), for all n in the stated confidence range.

Thus (27) is at most α, which gives $G_\alpha \leq s_{\mathrm{B}}$. Proposition 9 gives $s_{\mathrm{P}} \geq s_{\mathrm{B}}$ throughout this range, proving the claim for both safeguarded conventions. □

The proposition is a pointwise identity statement for the pre-specified safeguard, not a conditional-coverage theorem for ordinary Stringer. The analytic proof has no finite sample-size cutoff. The repository retains the earlier 59,700 exact dyadic comparisons through $n = 200$ at 90%, 95%, and 99% as an independent regression, not as the logical basis of the result. Controlling the remaining cap regions is a separate open problem.

6 Exact counterexamples at low confidence

Theorem 1 covers every confidence level, so the asymptotic anti-conservatism at $\alpha > 1/2$ [15] must manifest itself only at larger n. Before giving explicit finite-sample counterexamples, we exclude the smallest possible nontrivial support.

Proposition 19 (One nonzero taint value)**.** *If F is supported on $\{0, v\}$ with $v \in (0, 1]$, then, for every n and α, $\mathbb{P}(\mathrm{SB} \geq \mu) \geq 1 - \alpha$.*

Proof. Let K denote the number of nonzero taints. Then $K \sim \mathrm{Bin}(n, q)$,

$$\mathrm{SB} = (1-v)p_n(0) + v\,p_n(K) \geq v\,p_n(K), \qquad \mu = qv.$$

The Clopper–Pearson coverage event $\{p_n(K) \geq q\}$ has probability at least $1 - \alpha$; on that event, $\mathrm{SB} \geq vq = \mu$. □

Each counterexample below is certified: its distributional parameters are exact dyadic rationals, its coverage is computed as an exact rational by full multinomial enumeration, and every comparison SB $\geq \mu$ is made with rational factor intervals. Their endpoints lie on a dyadic grid with denominator 2^{80}, and the sign of the binomial CDF at each endpoint is evaluated exactly with integer arithmetic. The table rows are regenerated from the version-2 certificate files by `summarize_certificates.py`.

confidence	n	smallest certified coverage	certified examples
30%	50	0.29815	3
32%	100	0.30956	9
35%	200	0.33729	1
35%	400	0.32630	1
37%	200	0.36079	6
37%	400	0.36777	13

Table 7: Certified finite-sample violations (binomial factors, three-point taint distributions with two nonzero taint values). The listed results illustrate the onset of failure in the examined parameter ranges.

A representative certificate has nominal confidence 30% ($\alpha = 0.7$) and $n = 50$. Set

$$\begin{aligned} v_1 &= 1, & q_1 &= \frac{8984394653678481}{2^{55}}, \\ v_2 &= \frac{8384891858230937}{2^{54}}, & q_2 &= \frac{5642023711000999}{2^{53}}, \\ & & q_0 &= \frac{4476307521281491}{2^{55}}. \end{aligned}$$

The complete distribution is $\mathbb{P}(T = v_1) = q_1$, $\mathbb{P}(T = v_2) = q_2$, and $\mathbb{P}(T = 0) = q_0$. Thus the two nonzero taint values are 1 and approximately 0.465455, and the corresponding probabilities are approximately 0.249367 and 0.626390. These parameters appear in the third record of the $n = 50$, $\alpha = 0.7$ certificate (`certified-alpha0.7-n50.json`). The certificate prints the exact rational coverage in full. Its decimal value is approximately 0.2988790828, below the nominal level 0.30. The minimum comparison gap certified by rational interval arithmetic is approximately 2.10×10^{-14}; each factor interval has width at most $2^{-80} \approx 8.27 \times 10^{-25}$. Pap and van Zuijlen [14] established non-conservatism for uniform-type taints through exact recursions; the finite-support examples here are readily reproducible and illustrate the onset of failure in the examined parameter ranges.

By contrast, searches at nominal confidence levels 50%, 90%, and 95% found no violation. Grid-plus-optimizer searches examined distributions with two nonzero taint values ($n \leq 100$ at 95% and 90%; $n \leq 300$ at 50%) and distributions with three nonzero taint values ($n \leq 25$ at 95%), in each case allowing mass at zero. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10}. Near-minimizers occurred as the largest taint approached 1, consistent with Proposition 19, the $\{0, 1\}$-support minimality result of [7], and the conjectured coverage infimum $1-\alpha$. For the Poisson factors, the smallest coverage found in the corresponding searches remained above nominal. At 95%, for example, the values were approximately 0.9716 ($n = 10$), 0.9573 ($n = 50$), and 0.9549 ($n = 100$), a positive numerical cushion that decreases over the listed sample sizes.

7 Reassessment of a previous finite-sample certification

Let

$$C_n(F) = \mathbb{P}_F\big(\mathrm{SB} \geq \mu(F)\big)$$

denote the coverage probability. In this section, $t_{1:n} \leq \cdots \leq t_{n:n}$ are the sample taints arranged in nondecreasing order, with $t_{0:n} = 0$ and $t_{n+1:n} = 1$. Let $q_n(i)$ be the lower one-sided Clopper–Pearson limit for i successes, with $q_n(0) = 0$. The lower and upper limits used here satisfy $q_n(i) = 1 - p_n(n-i)$.

Bimpeh [6, pp. 76–79] writes the Stringer bound as $\mathrm{SB} = \int_0^1 [1 - \widehat{F}_{n,L}(t)]\,dt$, where the stepwise band is defined by

$$\widehat{F}_{n,L}(t) = q_n(i-1) \quad \text{for } t_{i-1:n} \leq t < t_{i:n}, \qquad 1 \leq i \leq n+1.$$

Let $V_{1:n} \leq \cdots \leq V_{n:n}$ denote the order statistics of n independent $\mathrm{Unif}(0,1)$ random variables. Using the probability integral transform and the reversal symmetry of uniform order statistics, Bimpeh proposes the lower bound

$$C_n(F) \;\geq\; \bar{P}_n \;:=\; \mathbb{P}\big(V_{i:n} \leq p_n(i),\ 1 \leq i \leq n\big). \tag{33}$$

Bimpeh evaluates $\bar{P}_n$ using Bolshev's recursion for uniform order-statistic boundary-crossing probabilities [19, pp. 366–367]. Because the computed values satisfy $\bar{P}_n \geq 1 - \alpha$ for $n \leq 11$ when $\alpha = 0.05$, the thesis concludes that (2) holds in that range. It also notes that

$$\bar{P}_2 = 2\sqrt{1-\alpha} - (1-\alpha) \geq 1 - \alpha$$

for every α and claims a proof for $n \leq 2$.

Proposition 20. *Inequality* (33) *fails for both continuous and atomic distributions F.*

Proof. Let $\alpha = 0.05$ and let F be the mixture assigning probability 0.99 to a uniform distribution on $(0.99, 1)$ and probability 0.01 to a uniform distribution on $(0, 0.01)$, so $\mu = 0.9851$. For $n = 1$, $\mathrm{SB} = 0.95 + 0.05\,T_1$ and $\bar{P}_1 = 1$. Moreover, $\mathrm{SB} < \mu$ if and only if $T_1 < 0.702$, an event of probability 0.01. Thus $C_1(F) = 0.99 < 1 = \bar{P}_1$. For $n = 2$, $\bar{P}_2 = 2\sqrt{0.95} - 0.95 \approx 0.99936$. A sample containing one taint from each mixture component fails, as does a sample containing two taints from the lower component. Therefore $C_2(F) = 1 - 2(0.99)(0.01) - 0.01^2 = 0.9801 < \bar{P}_2$.

For an atomic example at $n = 5$, let $\mathbb{P}(T = 1) = \frac{1}{2}$ and $\mathbb{P}(T = \frac{1}{10}) = \mathbb{P}(T = 0) = \frac{1}{4}$. Its exact coverage is $C_5(F) = \frac{31}{32} = 0.96875 < \bar{P}_5 \approx 0.98746$. The failure event is precisely the event of drawing no taint-1 unit, which has probability 2^{-5}. □

The issue is an index shift. Pointwise domination $F \geq \widehat{F}_{n,L}$ on $[0,1]$ requires $F(t_{i:n}) \geq q_n(i)$ for all i, because $q_n(i)$ is the band's value at the jump point $t_{i:n}$. The cited argument instead uses the left limit $q_n(i-1)$. In particular, it omits the constraint $F(t_{n:n}) \geq q_n(n) = \alpha^{1/n}$. Correcting the index does not recover the claimed coverage guarantee.

Proposition 21. *For every continuous F and every n, the corrected containment probability satisfies* $\mathbb{P}\big(F(t_{i:n}) \geq q_n(i)\ \forall i\big) = \mathbb{P}\big(V_{i:n} \geq q_n(i)\ \forall i\big) \leq 1-\alpha$.

Proof. For continuous F, the probability integral transform gives $F(t_{i:n}) = V_{i:n}$ in distribution. The corrected event includes its $i = n$ constraint, and $q_n(n) = \alpha^{1/n}$. Hence

$$\mathbb{P}\big(V_{i:n} \geq q_n(i)\ \forall i\big) \leq \mathbb{P}\big(V_{n:n} \geq \alpha^{1/n}\big) = 1-\alpha,$$

because $\mathbb{P}(V_{n:n} < x) = x^n$. □

Thus the corrected pointwise-containment event has probability no greater than nominal coverage. The integral inequality $\int \widehat{F}_{n,L} \leq \int F$ may hold on a larger event than pointwise domination, but the cited containment argument does not establish that fact. Our implementation reproduces the values of $\bar{P}_n$ and Bimpeh's Table 5.1 exactly; those values are boundary-crossing probabilities, not established coverage bounds. Consequently, the cited argument does not establish finite-sample conservatism at $n = 2$. Theorem 1 supplies that result directly.

8 Discussion

Three observations suggest directions for future work. First, the reformulation behind (3) generalizes: with $u_{(1)} \leq \cdots \leq u_{(n)}$ the increasing order statistics of $U_i = 1 - T_i$,

$$\mathrm{SB} \;=\; 1 - \sum_{j=1}^{n} e_j\, u_{(j)}, \qquad e_j = p_n(j) - p_n(j-1),$$

so (2) is a family of weighted order-statistic exceedance inequalities, with the audit-specific confidence-factor structure encoded in the positive increments e_j. Second, the $n = 2$ proof combines rectangle lower bounds on the covering event, a survival-integral budget, and a concave potential with equal endpoint values. The $n = 3$ through $n = 6$ proofs use a different route: they compare Stringer with a valid Dirichlet-average bound and certify the resulting simplex-cap inequalities. At $n = 6$ the exact computation certifies all three conventional levels. In higher dimensions, the analogous pointwise comparison is a concrete geometric problem for caps of a uniform simplex. Proposition 18 solves the complete one-upper-knot region at every sample size whenever nominal confidence is at least 75%. For the binomial factors at 90%, 95%, and 99%, $n = 7$ is the first sample size with unresolved cap regions. A dimension-free route recorded in the repository writes the Poisson bound as an ordered interpolation of gamma quantiles and reduces its pointwise Gaffke comparison to two high-quantile inequalities for weighted exponentials and uniform Dirichlet averages. Those inequalities remain conjectural. Third, the near-equality observed at every searched n, together with the certified failures at several nominal confidence levels below 50%, suggests that (2) may hold whenever nominal confidence is at least 50%, equivalently whenever $\alpha \leq 1/2$, with coverage infimum $1-\alpha$ at every n. This is a concrete conjecture for future work.

From an auditing perspective, the binomial Stringer bound has no uniform safety margin above nominal coverage: at $n = 2$, its coverage infimum is $1-\alpha$. Theorems 5, 6, and 7 supply exact guarantees at the three conventional levels for $n = 3$, $n = 4$, and $n = 5$; Theorem 8 adds all three guarantees at $n = 6$. At conventional confidence levels, the Poisson version pointwise dominates the binomial version by Proposition 9. The corrected simultaneous-band argument in Theorem 12 extends the rigorous Poisson guarantees to every $n \leq 8$ at 90%, every $n \leq 11$ at 95%, and every $n \leq 20$ at 99% confidence; the reported Poisson searches at larger sizes retain a positive but decreasing numerical cushion. For those larger sizes, Theorem 13 supplies a second distribution-free option that stays within the Poisson Stringer factor family: multiply the complete result by the precomputed $\kappa_{n,\alpha}$ and cap at one, or use the pointwise no-larger calibrated-factor-capped curve. This is a guarantee for the calibrated rule, not evidence about coverage of the ordinary rule. Corollary 14 gives a closed-form alternative: the multiplier two works whenever nominal confidence is at least $2/e$. Corollary 15 retains the first 100 crossing probabilities and reduces the uniform choices to 1.53, 1.44, and 1.33 at 90%, 95%, and 99%, respectively. The exact joint calibration is appreciably smaller in the reported examples. Corollary 16 gives a second calibration path that preserves the ordinary all-zero-sample result and scales only the error increments. The path must be chosen before the sample is observed. The certified examples show that anti-conservatism can occur at several nominal confidence levels below 50%, at sample sizes $n = 50, 100, 200$, and 400. Coverage for ordinary audit sample sizes at conventional confidence remains an open problem, so the search evidence alone does not justify a general validity claim for ordinary Stringer. Proposition 17, however, supplies an all-sample-size distribution-free reporting option under the model of this paper: pre-specify the maximum of the familiar Stringer calculation and the validated Gaffke floor. This fallback requires professional-standards and workflow evaluation before practice adoption, but it does not require the unresolved Stringer conjecture. Proposition 18 additionally proves that the fallback has zero uplift at every sample size and nominal confidence level of at least 75% whenever binomial Stringer is at least the largest observed taint.

Reproducibility

The repository contains exact rational coverage calculations, margin certificates, exact-sign dyadic enclosures for the binomial confidence factors, exact Bernstein certificates for Theorems 5, 6, 7, and 8, symbolic regeneration checks for their simplex-cap formulas, exact rational Poisson-limit and boundary-crossing certificates for Theorem 12, exact rational opposite-endpoint certificates for the representative full-scale and zero-taint-preserving calibrations in Theorem 13 and Corollary 16, exact rational exponential enclosures for both uniform-multiplier criteria in Table 5, high-precision Poisson search calculations, exact-sign rational brackets for the all-sample-size Gaffke safeguard, symbolic and rational checks for the analytic all-n one-cap comparison, an independent finite regression containing 59,700 exact dyadic comparisons through $n = 200$, search logs, symbolic checks of the identities used in the $n = 2$ proof, and stress tests over approximately 200,000 distributions. It also contains records of two separate adversarial review passes.

The canonical source for this development revision is github.com/RichieSater/stringer-bound. The earlier release v1.0.0 is preserved on Zenodo at `https://doi.org/10.5281/zenodo.21850820`; the version-independent DOI, which resolves to the latest archived release, is `https://doi.org/10.5281/zenodo.21850819`.

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